Digital Communications (ECEN5692) Midterm Practice Examination Solution

3/12/2026

Instructions: Four sides of letter size notes and calculators are allowed.
Time: 75 mins

Name:

Grade (Full score=130):

Problem 1. (30 pts)

Find $x_I(t)$, $x_Q(t)$ of the baseband signal $x(t)$ that produces the following passband signal $x_c(t)$.

1. (5/10 pts) $x_c(t) = \sqrt{8}m(t)\cos\left(2\pi f_c t + \frac{\pi}{4}\right)$, where $m(t)$ is a real-valued baseband message signal.
2. (5/10 pts) $x_c(t) = 4\cos(2\pi(f_c + f_\Delta)t)$, where f_Δ is a constant frequency offset ($f_\Delta \ll f_c$).

Solution:

We use the passband/baseband relation

$$x_c(t) = \Re\left\{\sqrt{2}\,x(t)e^{j2\pi f_c t}\right\}, \qquad x(t) = x_I(t) + jx_Q(t).$$

Also,

$$\cos(x) = \Re\{e^{jx}\}, \qquad \sin(x) = \Re\{-je^{jx}\}.$$

The idea is to rewrite each passband signal in the form

$$x_c(t) = \Re\left\{Ce^{j2\pi f_c t}\right\},$$

and then identify

$$C = \sqrt{2}\,x(t).$$

1. We can solve this by converting the cosine function into its complex exponential representation using the identity $\cos(\theta) = \Re\left[e^{j\theta}\right]$:

   $$x_c(t) = \sqrt{8}m(t)\Re\left[e^{j\left(2\pi f_c t + \frac{\pi}{4}\right)}\right]$$

   Since $\sqrt{8}m(t)$ is real-valued, we can bring it inside the real-part operator and separate the exponential terms:

   $$x_c(t) = \Re\left[\sqrt{8}m(t)e^{j\frac{\pi}{4}}e^{j2\pi f_c t}\right]$$

   We need to match this to the standard form:

   $$x_c(t) = \Re\left[\sqrt{2}x(t)e^{j2\pi f_c t}\right]$$

   Equating the complex envelopes:

   $$\sqrt{2}x(t) = \sqrt{8}m(t)e^{j\frac{\pi}{4}} \implies x(t) = \frac{\sqrt{8}}{\sqrt{2}}m(t)e^{j\frac{\pi}{4}} = 2m(t)e^{j\frac{\pi}{4}}$$

   Expanding the complex exponential using Euler's formula $e^{j\frac{\pi}{4}} = \cos\left(\frac{\pi}{4}\right) + j\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}$:

   $$x(t) = 2m(t)\left(\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}\right) = \sqrt{2}m(t) + j\sqrt{2}m(t)$$

Because $x(t) = x_I(t) + jx_Q(t)$, we can easily identify the in-phase and quadrature components:

$$x_I(t) = \sqrt{2}m(t)$$

$$x_Q(t) = \sqrt{2}m(t)$$

Therefore, $x_I(t) = \sqrt{2}m(t)$ and $x_Q(t) = \sqrt{2}m(t)$.

2. We can solve this easily by converting the signal into its complex exponential representation.

$$x_c(t) = \Re\left[4e^{j2\pi(f_c+f_\Delta)t}\right] = \Re\left[4e^{j2\pi f_\Delta t}e^{j2\pi f_c t}\right]$$

We need to match this to the standard form:

$$x_c(t) = \Re\left[\sqrt{2}x(t)e^{j2\pi f_c t}\right]$$

Equating the complex envelopes:

$$\sqrt{2}x(t) = 4e^{j2\pi f_\Delta t} \implies x(t) = \frac{4}{\sqrt{2}}e^{j2\pi f_\Delta t} = 2\sqrt{2}e^{j2\pi f_\Delta t}$$

Expanding this using Euler's formula gives the in-phase and quadrature components:

$$x(t) = 2\sqrt{2}\left[\cos(2\pi f_\Delta t) + j\sin(2\pi f_\Delta t)\right]$$

Because $x(t) = x_I(t) + jx_Q(t)$:

$$x_I(t) = 2\sqrt{2}\cos(2\pi f_\Delta t)$$

$$x_Q(t) = 2\sqrt{2}\sin(2\pi f_\Delta t)$$

Therefore, $x_I(t) = 2\sqrt{2}\cos(2\pi f_\Delta t)$ and $x_Q(t) = 2\sqrt{2}\sin(2\pi f_\Delta t)$.

Problem 2. (40 pts)

A zero-mean, white, real, stationary, and Gaussian noise process $\boldsymbol{w}(t)$ with $r_{\boldsymbol{w}}(\tau) = \frac{N_0}{2}\delta(\tau)$ is the input to a passband filter $\tilde{h}_c(f)$ and the output is $\boldsymbol{n}_c(t)$. The passband noise $\boldsymbol{n}_c(t)$ is the input to an I/Q down converter, which produces output $\boldsymbol{n}(t) = \boldsymbol{n}_I(t) + j\boldsymbol{n}_Q(t)$. We know $f_c = 1000$Hz and the filter has an asymmetric frequency response:

$$\tilde{h}_c(f) = \begin{cases} \sqrt{2} & 500 \leq |f| \leq 1000 \\ \sqrt{4 - \frac{|f|}{500}} & 1000 < |f| \leq 2000 \\ 0 & \text{elsewhere} \end{cases}.$$

1. (2/10 pts) Calculate $\tilde{r}_{\boldsymbol{n}_c}(f)$, the power spectral density of $\boldsymbol{n}_c(t)$. Draw the picture.
2. (2/10 pts) Calculate $\tilde{r}_{\boldsymbol{n}}(f)$, the complex baseband power spectral density of $\boldsymbol{n}(t)$. Draw the picture.
3. (2/10 pts) Calculate $\tilde{r}_{\boldsymbol{n}_I}(f)$, the PSD of the in-phase noise component.
4. (2/10 pts) Calculate the cross-power spectral density $\tilde{r}_{\boldsymbol{n}_I,\boldsymbol{n}_Q}(f)$ and determine whether the processes $\boldsymbol{n}_I(t)$ and $\boldsymbol{n}_Q(t)$ are independent. Justify your answer.
5. (2/10 pts) Calculate $\sigma^2_{\boldsymbol{n}_I(3)}$, the variance of $\boldsymbol{n}_I(3)$.

Solution:

1.

$$\begin{aligned} \tilde{r}_{\boldsymbol{n}_c}(f) &= \tilde{r}_{\boldsymbol{w}}(f)|\tilde{h}_c(f)|^2 \\ &= \begin{cases} N_0 & 500 \leq |f| \leq 1000 \\ N_0\left(2 - \frac{|f|}{1000}\right) & 1000 < |f| \leq 2000 \\ 0 & \text{elsewhere} \end{cases}. \end{aligned}$$

2. Because the complex baseband PSD is related to the passband PSD via

$$\tilde{r}_{\boldsymbol{n}_c}(f) = \frac{\tilde{r}_{\boldsymbol{n}}(f - f_c) + \tilde{r}_{\boldsymbol{n}}(-f - f_c)}{2},$$

or $\tilde{r}_{\boldsymbol{n}}(f) = 2\tilde{r}_{\boldsymbol{n}_c}(f + f_c)|\tilde{h}_L(f)|^2$, we obtain

$$\tilde{r}_{\boldsymbol{n}}(f) = \begin{cases} 2N_0 & -500 \leq f \leq 0 \\ 2N_0\left(1 - \frac{f}{1000}\right) & 0 < f \leq 1000 \\ 0 & \text{elsewhere} \end{cases}.$$

3. The PSD of the in-phase component is given by the even part of $\tilde{r}_{\boldsymbol{n}}(f)$ scaled by 1/2. In formula:

$$\begin{aligned}
\tilde{r}_{\boldsymbol{n}_I}(f) &= \frac{\tilde{r}_{\boldsymbol{n}}(f) + \tilde{r}_{\boldsymbol{n}}(-f)}{4} \\
&= \begin{cases} \frac{1}{4}\left[2N_0\left(1 - \frac{|f|}{1000}\right) + 2N_0\right] = \frac{N_0}{2}\left(2 - \frac{|f|}{1000}\right) & |f| \leq 500 \\ \frac{1}{4}\left[2N_0\left(1 - \frac{|f|}{1000}\right) + 0\right] = \frac{N_0}{2}\left(1 - \frac{|f|}{1000}\right) & 500 < |f| \leq 1000 \\ 0 & \text{elsewhere} \end{cases} .
\end{aligned}$$

4. The cross-PSD is related to the odd part of $\tilde{r}_{\boldsymbol{n}}(f)$:

$$\tilde{r}_{\boldsymbol{n}_I,\boldsymbol{n}_Q}(f) = j\frac{\tilde{r}_{\boldsymbol{n}}(f) - \tilde{r}_{\boldsymbol{n}}(-f)}{4}$$

For $f > 0$:

$$\tilde{r}_{\boldsymbol{n}_I,\boldsymbol{n}_Q}(f) = \begin{cases} \frac{j}{4}\left[2N_0\left(1 - \frac{f}{1000}\right) - 2N_0\right] = -j\frac{N_0 f}{2000} & 0 < f \leq 500 \\ \frac{j}{4}\left[2N_0\left(1 - \frac{f}{1000}\right) - 0\right] = j\frac{N_0}{2}\left(1 - \frac{f}{1000}\right) & 500 < f \leq 1000 \\ -j\frac{N_0}{2}\left(1 + \frac{f}{1000}\right) + j\frac{N_0}{2} = -j\frac{N_0 f}{2000} & -500 \leq f \leq 0 \\ -j\frac{N_0}{2}\left(1 + \frac{f}{1000}\right) & -1000 \leq f < -500 \\ 0 & \text{elsewhere} \end{cases} .$$

5. Since the noise process is Wide-Sense Stationary (WSS), the variance is time-invariant (i.e., independent of $t = 3$). We can find the variance by

integrating the PSD of the in-phase component over all frequencies:

$$
\begin{aligned}
\sigma^2_{\boldsymbol{n}_I(3)} &= r_{\boldsymbol{n}_I}(0) \\
&= \int_{-\infty}^{\infty} \tilde{r}_{\boldsymbol{n}_I}(f)df \\
&= 2\int_{0}^{1000} \tilde{r}_{\boldsymbol{n}_I}(f)df \quad \text{(since the PSD is an even function)} \\
&= 2\left[\int_{0}^{500} N_0\left(1-\frac{f}{2000}\right)df + \int_{500}^{1000}\frac{N_0}{2}\left(1-\frac{f}{1000}\right)df\right] \\
&= 2\left[N_0\left(f-\frac{f^2}{4000}\right)\Bigg|_0^{500} + \frac{N_0}{2}\left(f-\frac{f^2}{2000}\right)\Bigg|_{500}^{1000}\right] \\
&= 2\left[N_0\left(500-\frac{250000}{4000}\right) + \frac{N_0}{2}\left(\left(1000-\frac{1000000}{2000}\right)-\left(500-\frac{250000}{2000}\right)\right)\right] \\
&= 2\left[N_0(500-62.5) + \frac{N_0}{2}\left((1000-500)-(500-125)\right)\right] \\
&= 2\left[437.5N_0 + \frac{N_0}{2}(500-375)\right] \\
&= 2\left[437.5N_0 + 62.5N_0\right] \\
&= 2(500N_0) \\
&= 1000N_0.
\end{aligned}
$$

Problem 3. (60 pts)

(*Two ray fading channel model*) A UAV (Unmanned Aerial Vehicle) sends a signal to a ground vehicle, as shown in Fig. 1.

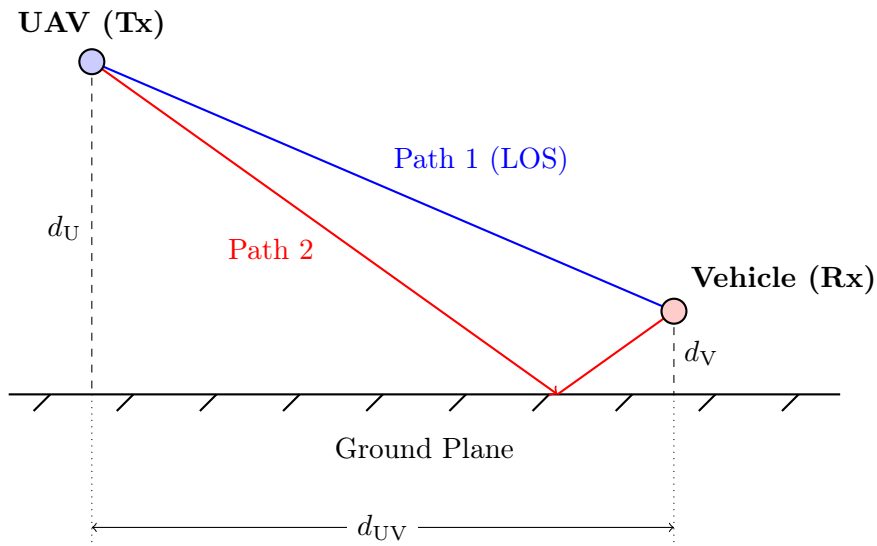

Figure 1: Two ray fading channel model (UAV to Vehicle)

Ray/path 1 is a line of sight path. Ray/path 2 is reflected from the ground. For simplicity, assume the ground is a perfect conductor adding a phase of π to the signal. If the baseband transmit signal is $x(t)$, the received baseband signal is

$$y(t) = \sum_{i=1}^{2} a_i e^{-j2\pi f_c \tau_i(t)} x(t - \tau_i(t)),$$

where $\tau_i(t)$ is the delay of path i; $a_i(t) = \frac{\beta}{r_i(t)} e^{-j\varphi_i}$, $i = 1,\ 2$; $\beta \in \mathbb{C}$ is a constant; $r_i(t)$ is the path length; $\varphi_1 = 0$, $\varphi_2 = \pi$. Using basic geometry of similar triangles, we can find

$$r_1(t) = \sqrt{d_{\mathrm{UV}}^2 + (d_{\mathrm{U}} - d_{\mathrm{V}})^2},$$

$$\begin{aligned} r_2(t) &= \sqrt{\left(\frac{d_{\mathrm{UV}} d_{\mathrm{U}}}{d_{\mathrm{U}} + d_{\mathrm{V}}}\right)^2 + d_{\mathrm{U}}^2} + \sqrt{\left(\frac{d_{\mathrm{UV}} d_{\mathrm{V}}}{d_{\mathrm{U}} + d_{\mathrm{V}}}\right)^2 + d_{\mathrm{V}}^2} \\ &= \sqrt{d_{\mathrm{UV}}^2 + (d_{\mathrm{U}} + d_{\mathrm{V}})^2}. \end{aligned}$$

Assume

$$\begin{aligned} f_{\mathrm{c}} &= 2.4\mathrm{GHz}, \\ d_{\mathrm{U}} &= 50\mathrm{m}, \\ d_{\mathrm{V}} &= 2\mathrm{m}, \\ d_{\mathrm{UV}} &= d_0 + vt, \\ d_0 &= 200\mathrm{m}. \end{aligned}$$

1. Assume nothing is moving, i.e., $v = 0$, and thus,

$$y(t) = h(t) * x(t).$$

   (a) (1/10 pts) Calculate $\tau_1(t) = \tau_1$ and $\tau_2(t) = \tau_2$.

   (b) (2/10 pts) Calculate the $h(t)$.

   (c) (2/10 pts) Calculate the $\tilde{h}(f)$.

2. Assume $v = 30$ m/s.

   (a) (2/10 pts) Calculate the path delay spread $\Delta\tau$ at time $t = 0$. Calculate the Doppler frequency $f_{\mathrm{d},1}$, $f_{\mathrm{d},2}$ and Doppler spread Δf_{d} at $t = 0$.

   (b) (3/10 pts) Assume the symbol period $= \frac{1}{\text{signal bandwidth}} \gg$ delay spread, and thus

$$y(t) \approx h(t)x(t - \tau_1(0)).$$

   Calculate $h(t)$.

Solution:

1. (a) Since nothing is moving, $\tau_i(t) = \tau_i$.

$$\begin{aligned} \tau_1 &= \frac{r_1}{c} \\ &= \frac{\sqrt{d_{UV}^2 + (d_U - d_V)^2}}{c} \end{aligned}$$

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=sqrt(200^2+(50-2)^2)/(3*10^8)
=6.8559787e-07
```

$$\begin{aligned} \tau_2 &= \frac{r_2}{c} \\ &= \frac{\sqrt{d_{UV}^2 + (d_U + d_V)^2}}{c} \end{aligned}$$

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=sqrt(200^2+(50+2)^2)/(3*10^8)
=6.8883154e-07
```

(b) Observe that

$$\begin{aligned} y(t) &= \sum_{i=1}^{2} a_i e^{-j2\pi f_c \tau_i} x(t-\tau_i) \\ &= \left(\sum_{i=1}^{2} a_i e^{-j2\pi f_c \tau_i} \delta(t-\tau_i) \right) * x(t). \end{aligned}$$

Consequently,

$$\begin{aligned} h(t) &= \sum_{i=1}^{2} a_i e^{-j2\pi f_c \tau_i} \delta(t-\tau_i) \\ &= \frac{\beta}{r_1} e^{-j2\pi f_c \tau_1} \delta(t-\tau_1) + \frac{\beta}{r_2} e^{j\pi} e^{-j2\pi f_c \tau_2} \delta(t-\tau_2) \\ &= \frac{\beta}{205.679} e^{-j2\pi 1645.4349} \delta(t - 6.8559787 \times 10^{-7}) - \\ &\quad \frac{\beta}{206.649} e^{-j2\pi 1653.1957} \delta(t - 6.8883154 \times 10^{-7}), \end{aligned}$$

where

```
r1=sqrt(200^2+(50-2)^2)
=2.0567936e+02
r2=sqrt(200^2+(50+2)^2)
=2.0664946e+02
```

(c)

$$\begin{aligned} \tilde{h}(f) &= \mathcal{F}\{h(t)\} \\ &= \sum_{i=1}^{2} a_i e^{-j2\pi f_c \tau_i} e^{-j2\pi f \tau_i} \\ &= \frac{\beta}{205.679} e^{-j2\pi 1645.4349} e^{-j2\pi f \tau_1} - \frac{\beta}{206.649} e^{-j2\pi 1653.1957} e^{-j2\pi f \tau_2} \\ &= e^{-j2\pi f \tau_1} \left(\frac{\beta}{205.679} e^{-j2\pi 1645.4349} - \frac{\beta}{206.649} e^{-j2\pi 1653.1957} e^{-j2\pi f (\tau_2 - \tau_1)} \right). \end{aligned}$$

2.

(a)

$$\begin{aligned} \Delta\tau &= |\tau_1(0) - \tau_2(0)| \\ &= \frac{|r_1(t) - r_2(t)|}{c}\Big|_{t=0} \\ &= \frac{\left| \sqrt{d_{UV}^2 + (d_U - d_V)^2} - \sqrt{d_{UV}^2 + (d_U + d_V)^2} \right|}{c}\Big|_{t=0} \end{aligned}$$

```
=(sqrt(200^2+(50+2)^2)-sqrt(200^2+(50-2)^2))/(3*10^8)
=3.233665e-09
```

$$
\begin{aligned}
f_{\mathrm{d},1} &= -f_{\mathrm{c}}\tau_1'(0) \\
&= -f_{\mathrm{c}}\frac{r_1'(0)}{c} \\
&= -f_{\mathrm{c}}\frac{v d_{\mathrm{UV}}(0)\left(d_{\mathrm{UV}}^2(0)+(d_{\mathrm{U}}-d_{\mathrm{V}})^2\right)^{-\frac{1}{2}}}{c}
\end{aligned}
$$

```
=-2.4*10^9*30*200*(200^2+(50-2)^2)^(-0.5)/(3*10^8)
=-233.3729
```

$$
\begin{aligned}
f_{\mathrm{d},2} &= -f_{\mathrm{c}}\tau_2'(0) \\
&= -f_{\mathrm{c}}\frac{r_2'(0)}{c} \\
&= -f_{\mathrm{c}}\frac{v d_{\mathrm{UV}}(0)\left(d_{\mathrm{UV}}^2(0)+(d_{\mathrm{U}}+d_{\mathrm{V}})^2\right)^{-\frac{1}{2}}}{c}
\end{aligned}
$$

```
=-2.4*10^9*30*200*(200^2+(50+2)^2)^(-0.5)/(3*10^8)
=-232.2774
```

$$
\Delta f_{\mathrm{d}} = |f_{\mathrm{d},1} - f_{\mathrm{d},2}|
$$

```
=abs(-233.3729-(-232.2774))
=1.0955
```

(b) Observe that

$$
\begin{aligned}
y(t) &= \sum_{i=1}^{2} a_i e^{-j2\pi f_{\mathrm{c}}\tau_i(t)} x(t-\tau_i(t)) \\
&\approx \left(\sum_{i=1}^{2} a_i e^{-j2\pi f_{\mathrm{c}}\tau_i(t)}\right) x(t-\tau_1(0)).
\end{aligned}
$$

Consequently,

$$
\begin{aligned}
h(t) &= \left(\sum_{i=1}^{2} a_i e^{-j2\pi f_{\mathrm{c}} \tau_i(t)}\right) \\
&\approx \left(\sum_{i=1}^{2} a_i e^{-j2\pi f_{\mathrm{c}} \tau_i(0)} e^{j2\pi f_{\mathrm{d},i} t}\right) \\
&= \frac{\beta}{205.679} e^{-j2\pi 1645.4349} e^{-j2\pi 233.3729t} - \\
&\quad \frac{\beta}{206.649} e^{-j2\pi 1653.1957} e^{-j2\pi 232.2774t}.
\end{aligned}
$$